CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY

Lecture No. 9

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TOPICS TO BE COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

Relations

$R_1 \to$ Transitive $\qquad R_2 \to$ Transitive

$R_1 = \{(1\,3)\}$ ✓ $\qquad R_2 = \{(3\,1)\}$ ✓

$(a,b) \in R \wedge (b,c) \in R \to (a,c) \in R$

$(1,3) \in R \wedge \boxed{(3,\ \ )}$

$a = 1$ $\qquad$ F

$b = 3$

F $\longrightarrow$

$R_1 \cup R_2 = \{(1\,3)(3\,1)\}$

$R_1 \cap R_2 = \phi$

Relations

	$\cup$	$\cap$
R_1, R_2 Transitive	✗	✓

Relations

$A \to$ non empty set

$\begin{cases} A \to \phi \\ R \to \phi \end{cases}$

$R = \phi$

Symmetric ✓

Asymmetric ✓

Antisymmetric ✓

Transitive ✓

Irreflexive ✓

Reflexive ✗

Relations

	Reflexive	Symmetric	Anti symmetric	Asym	Irreflexive
Reflexive	—	$2^{n^2-n/2}$	$3^{n^2-n/2}$	0	0
Symme		—	2^{n}	$1(\phi)$	$2^{n^2-n/2}$
Anti			—	$3^{n^2-n/2}$	$3^{n^2-n/2}$
Asy				—	$3^{n^2-n/2}$
Irr.					—

$R = \{ \}$

$R \rightarrow 2^{n^2-n}$

$Irr \rightarrow 2^{n^2-n}$

$Sym \rightarrow 2^n \cdot 2^{\frac{n^2-n}{2}}$

$\underline{Anti}: 2^n \cdot 3^{n^2-n/2}$

$Asy \rightarrow 3^{n^2-n/2}$.

$R \cap \underline{Sy}: 2^{n^2-n/2}$

Relations

<u>Anti</u>

$$\underbrace{2^n}_{\text{Diagonal}} \cdot \underbrace{3^{n^2-n/2}}_{\text{boxes}}$$

Asy ∩ Anti

* no same elements ∩ same elements.
* no flipping ∩ no flipping

$= \text{no flipping}$

$= 3^{n^2-n/2}$

Relations

$R_1 = \{ (a, b) \mid a \leq b \}$

$R_2 = \{ (e_1, e_2) \mid e_1 R e_2 \}$

$R = \{ (G_1, G_2) \mid G_1 \equiv \bar{G_1} \}$

$R = \{ (set1, set2) \mid set1 \subseteq set2 \}$

$R = \{ (H_1, H_2) \mid H_1 \text{ is brother of } H_2 \}$

Relations

$R = \{ (A, B) \mid A \subseteq B \}$

R: $A \subseteq A$ ✓

Sy: $A R B \rightarrow B R A$

$A \subseteq B \rightarrow B \subseteq A$ (false)

$\{1\} \subseteq \{1, 3\} \rightarrow \{1, 3\} \subseteq \{1\}$

$a_1 \, R \, a_2$

$(1, A, A_{11}) \; R \; (2, B, B_{11})$

$(3, A, A_{11}, 99..) \; R \; (4, B, B_{11}, 88..)$

id	name	add	mobile
3	A	A11	99
4	B	B11	88

Relations

Z :

$z^2 = \{ (1,3), (\ ), (\ ), (\ ) \dots \}$

$R = \{ (a,b) \mid a R b \}$

set: z^2

CP: $z^2 \times z^2$.

$\{ (\overset{o1}{(\ )}, \overset{o2}{(\ )}), ((\ ), (\ )) \}$

$R = \{ (o1, o2) \mid o1\, R\, o2 \}$

Relations

Set: Z^2

CP. $Z^2 \times Z^2 = \{((\ ),(\ ))\ ((\ ),(\ ))\ \ldots$

$R = \{ (a,b)\, R\, (c,d) \mid \underline{ad = bc} \}$

<u>Symmetric:</u>

<u>Reflexive:</u> $(a,b)\, R\, (a,b)$ ✓

$a \cdot b = b \cdot a$

Relations

$R : \{ (\overset{1}{a}, \overset{2}{b}) \, R \, (\overset{3}{c}, \overset{4}{d}) \mid ad = bc \}$

$1 \cdot 4 = 2 \cdot 3$

$aRb \rightarrow bRa.$

Symmetric:

$(\overset{1}{a}, \overset{2}{b}) \, R \, (\overset{3}{c}, \overset{4}{d}) \longrightarrow (\overset{1}{c}, \overset{2}{d}) \, R \, (\overset{3}{a}, \overset{4}{b})$

$ad = bc \longrightarrow c \cdot b = d \cdot a.$

$bc = ad.$

$ad = bc$

Relations

$$R = \{(a, b) R (c, d) \mid ad = bc\}$$

$ad = bc \wedge cy = dx$

$\frac{cy}{x} = d$

$a \cdot \frac{\cancel{c} y}{x} = b \cdot \cancel{c}$

$ay = bx.$

<u>Transitive</u>:

$$(a, b) R (c, d) \wedge (c, d) R (x, y) \longrightarrow (a, b) R (x, y)$$

$ad = bc \quad \wedge \quad cy = dx$

$\frac{cy}{x} = d$

$a \cdot \frac{cy}{x} = bc$

$\underline{ay = bx.}$

Relations

(GATE-15)

$$R : \left\{ (\overset{1}{p}, \overset{2}{q})\, R\, (\overset{3}{r}, \overset{4}{s}) \;\middle|\; p - s = q - r \right\}$$

$1 - 4 = 2 - 3$

Reflexive. ?

Symmetric. ?

Reflexive. $(\overset{1}{p}, \overset{2}{s})\, R\, (\overset{3}{p}, \overset{4}{s})$

$1 - 4 = 2 - 3$

$p - s = s - p$

✗.

Symmetric:

$(a, b)\, R\, (c, d) \rightarrow (c, d)\, R\, (a, b)$

mistake.

$a - d = b - c$

$c - b = d - a$

$-(b - c) = -(a - d)$

$b - c = a - d.$

Relations

$R: \left\{ \underset{1}{(a}, \underset{2}{b}) R (\underset{3}{c}, \underset{4}{d}) \;\middle|\; a \le c \text{ OR } b \le d \right\}$

H.W: Reflexive. Transitive

$R = \left\{ (a, b) R (c, d) \;\middle|\; a^2 + d^2 = b^2 + c^2 \right\}$ R / Sy / Tra.

Relations

<u>equivalence Relation</u> :

all @ same time
- Reflexive
- Symmetric
- Transitive

$R = \{ \underline{(1\,1)(2\,2)(3\,3)}\ \underline{(1\,3)(3\,1)} \}$

R ✓

Sy. ✓

T. ✓

Relations

$a, b \in z.$

$R = \{ (a, b) \mid a+b = \text{even} \}$ equivalence Relⁿ

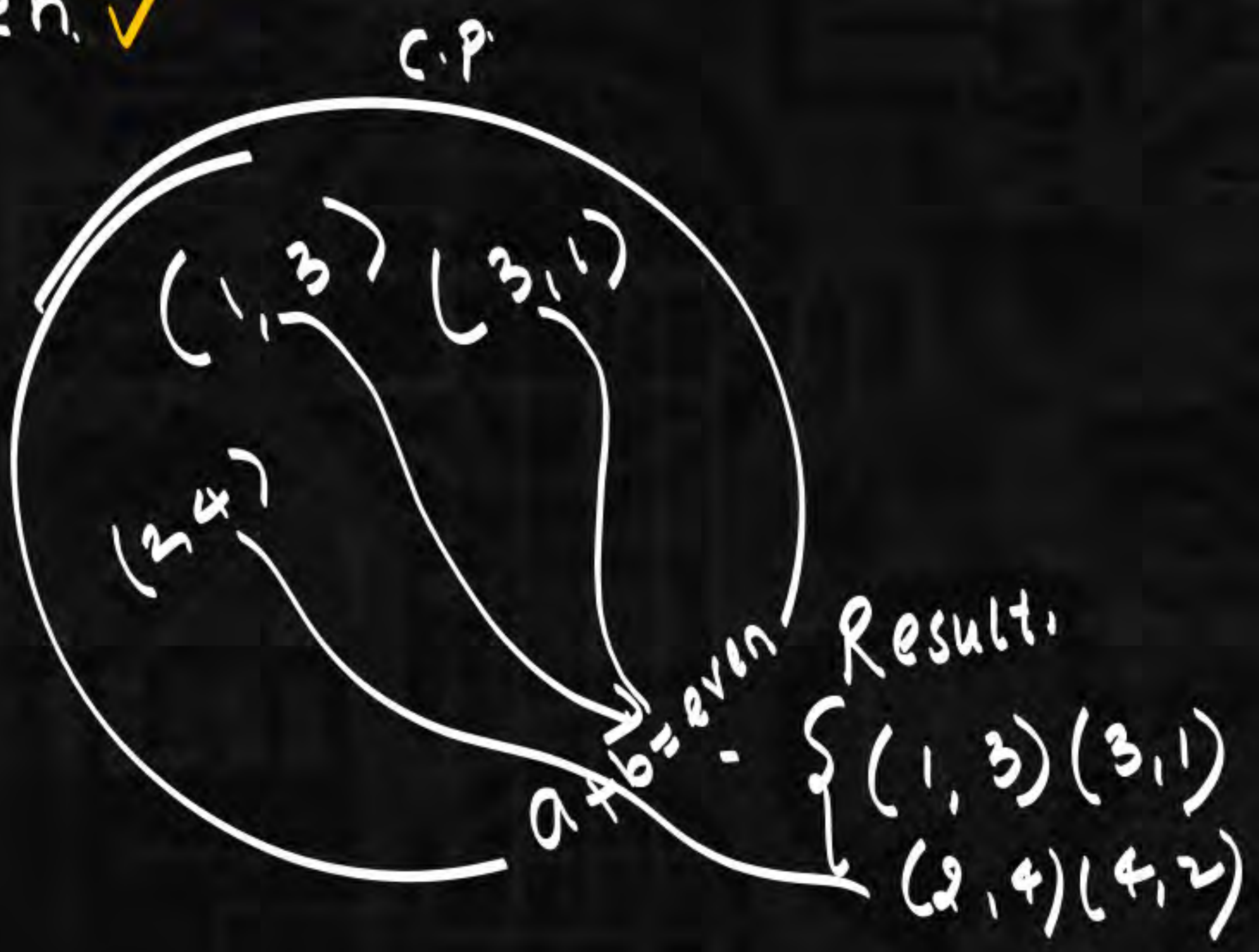
$2 \cdot \_ = \text{even}$

$\underline{R}$: aRa $\quad a+a = \text{even}. \quad 2a = \text{even}. \checkmark$

$\underline{Sy}$: $aRb \longrightarrow bRa$

$a+b = \text{even} \longrightarrow b+a = \text{even} \checkmark$

$\underline{(1,3) \in R} \longrightarrow (3,1) \in R$



Relations

$o + o = \text{even}$

$e + e = \text{even}.$

Transitive: $aRb \wedge bRc \longrightarrow aRc.$

$$\overset{o+o}{a+b=\text{even}} \wedge \overset{o+o}{b+c=\text{even}} \longrightarrow \overset{o+o}{a+c=\text{even}}.$$

$$(1,3) \in R \wedge (3,5) \in R \longrightarrow (1,5) \in R.$$

$$1+3=\text{even} \wedge 3+5=\text{even} \longrightarrow 1+5=\text{even}$$

Relations

$R = \{ (a,b) \mid a \le b \}$ → not equivalence Relt^n.

R: $aRa \quad a \le a$ ✓

Sy: $aRb \rightarrow bRa$ ✗

$a \le b \rightarrow b \le a$.

$\underset{T}{\underline{1 \le 3}} \longrightarrow \underset{F}{\underline{3 \le 1}}$

$(1,3) \in R \rightarrow (3,1) \in R$

$1 \le 3 \longrightarrow 3 \le 1 \ (F)$

$(1,3) \quad (3,1)$

$a \le b$

$R = \{ (1,3) \}$

Relations

$R = \{(a,b) \mid a \equiv b \pmod{n}\} \rightarrow$ equivalence Reltn.

$a \equiv b \pmod{n}$

a, b are having same remainder w.r.t (n)

R: aRa $\quad a \equiv a \pmod{n}$ ✓ $\quad 5 \equiv 5 \pmod{4}$

$1 \equiv 5 \pmod{4}$

Sy: $aRb \longrightarrow bRa$ ✓

$a \equiv b \pmod{n} \rightarrow b \equiv a \pmod{n}$

$1 \equiv 5 \pmod{4} \rightarrow 5 \equiv 1 \pmod{4}$

Tra:

$a \equiv b \pmod{n} \wedge b \equiv c \pmod{n}$

$\longrightarrow a \equiv c \pmod{n}$

$1 \equiv 5 \pmod{4} \wedge 5 \equiv 9 \pmod{4} \longrightarrow 1 \equiv 9 \pmod{4}$

Relations

$D: Z^+$

$\forall a \forall b \in Z^+$

$R = \{ (a, b) \mid a \text{ divides } b \}$

$a \mid b$

not equivalence Relation.

a divides b

OR

$\frac{b}{a}$

OR

$a \mid b$

R: aRa $\quad a \mid a$.

Symm: $aRb \to bRa$. ✗

$a \mid b \to b \mid a$

$3 \mid 6 \to 6 \nmid 3$

$(3,6) \in R \to (6,3) \notin R$

Transitive:

$a \mid b \wedge b \mid c \to a \mid c$

Relations

equivalence Relation creates partitions on a set.

↳ equivalence classes.

Set :

CP :

$\boxed{\text{Result: / Relation:}}$

equivalence Relation.

Relations

Set : Z

C.P.

$R : \{(a,b) \mid a+b = \text{even}\}$

equivalence Relt^n.

Set :

$\rightarrow$ 1, 3, 5, 7, 9

$\rightarrow$ 0, 2, 4, 6, 8, 10

$(1,3) \in R$

$(2,4) \in R$

$aRb \quad (a,b) \in R \rightarrow$ same partions

$xRy \quad (x,y) \in R \rightarrow$ same partion.

Relations

RST $\begin{cases} (1,1)(3,3)(5,5) \\ (1,3)(3,1)(1,5)(5,1) \end{cases}$

RST $\begin{cases} \\ \end{cases}$

1R3
1R5

$1+0 \neq$ even $1 \not R\, 0$

$1+2 \neq$ even $1 \not R\, 2$

<u>Set</u>:

1, 3, 5, 7, 9

0, 2, 4, 6, 8, 10

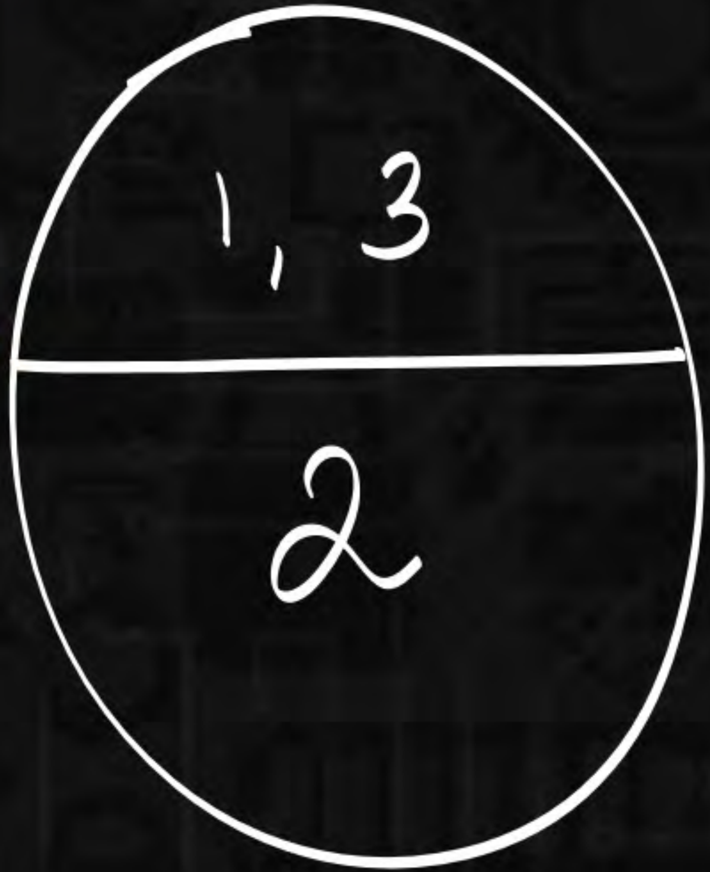
$(1,3) \in R$

$(2,4) \in R$

Relations

$R = \{(1\,1)(2\,2)(3\,3)(1\,3)(3\,1)\}$

$A = \{1, 2, 3\}$



Relations

$R = \{(a, b) \mid a \equiv b \pmod{4}\}$

D: 2.

$0 \equiv 4 \pmod 4$

$(0,0)$

$(0,1) \notin R$ $(0,2) \notin R$

$(0,4) \in R$

$(0,3) \notin R$

0, 4, 8
1, 5, 9
2, 6, 10
3, 7, 11

Relations

A → non empty set.

R → equivalence Relation.

1) $A_1 \cup A_2 \cup A_3 = A$.

2) $A_1 \cap A_2 \cap A_3 = \emptyset$

$[a_1]_R$ Representative of class A_1.

Relations

Total no. of equivalence Relations = Total no. of diff partitions

Relations

11 22 33 (13) (31) → 1, 3 | 2

11 22 33 (12) (21) → 1, 2 | 3

11 22 33 23 32 → 2, 3 | 1

S(3, 2)

{11 22 33} → 1 | 2 | 3

S(3,3)

A × A → 1, 2, 3

S(3, 1)

Relations

$f : A \rightarrow B$

1

2

3

I

II

$2^3 - 2 = 6$ onto

Rooms are same.

1

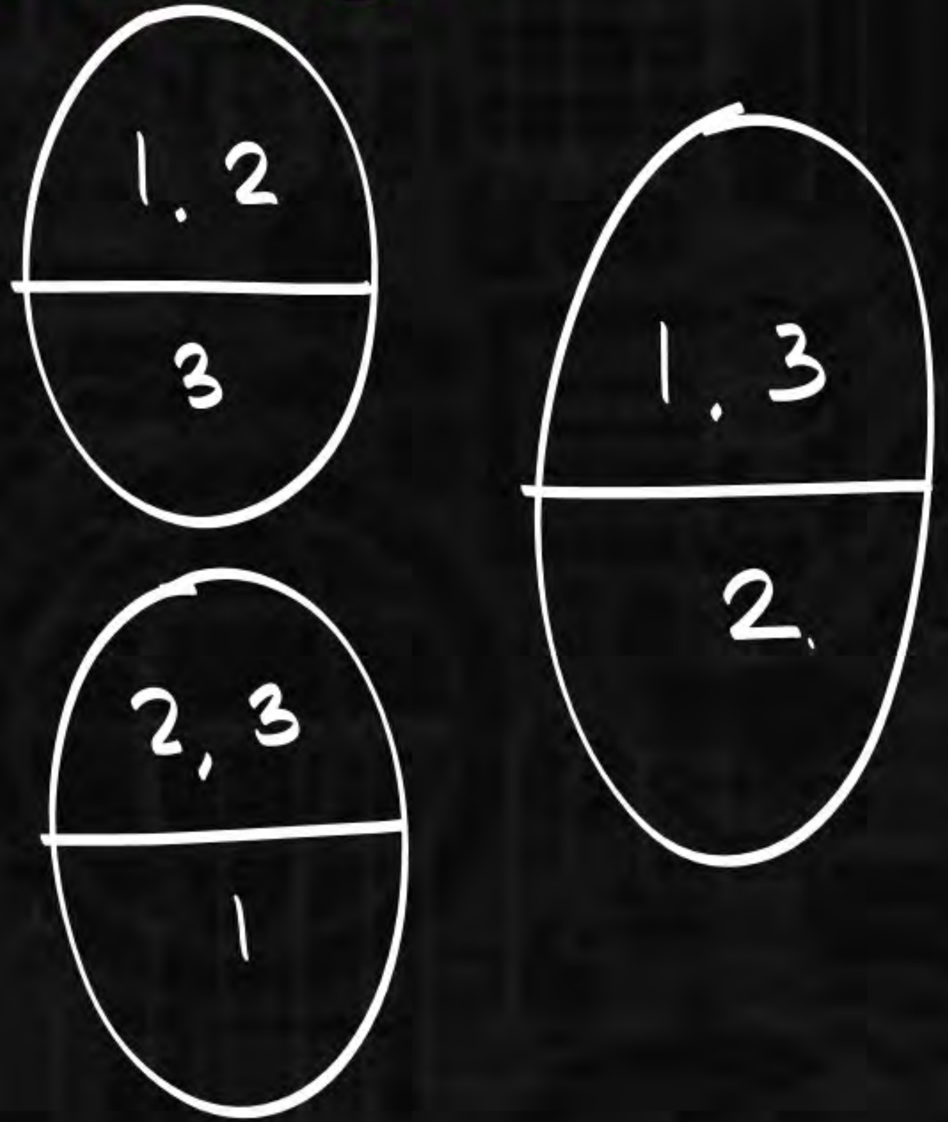
2

3

$\frac{6}{3!}$

$S(3,2) = 3$

$\frac{\text{onto}}{2!} = 6/2 = 3$



Relations

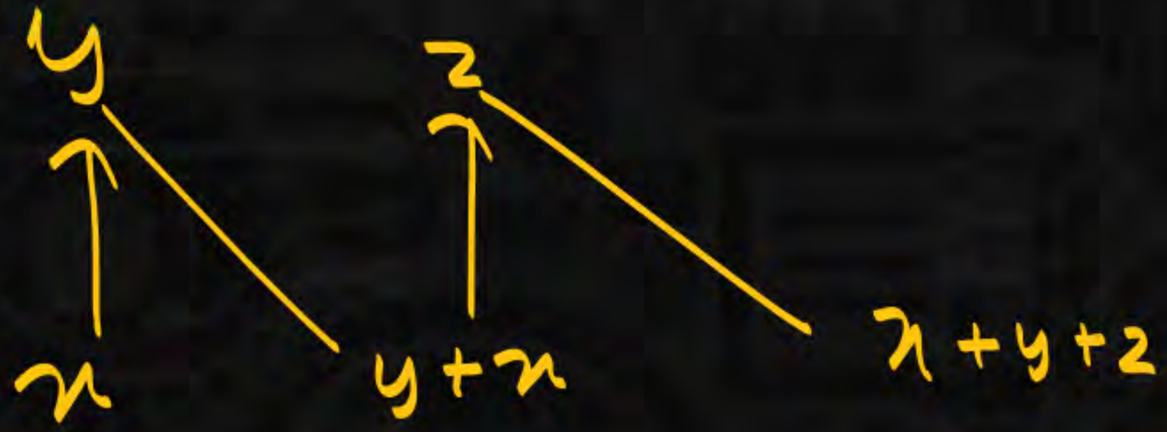
Total no. of equivalence relation

= Total no. of partitions = $\underline{S(3,1) + S(3,2) + S(3,3)}$

$$\underline{B(n)} = \sum_{i=0}^{k} S(n,k)$$

$\hookrightarrow$ $\underline{\text{bell no.}}$

Relations




THANK YOU GW
SOLDIERS !